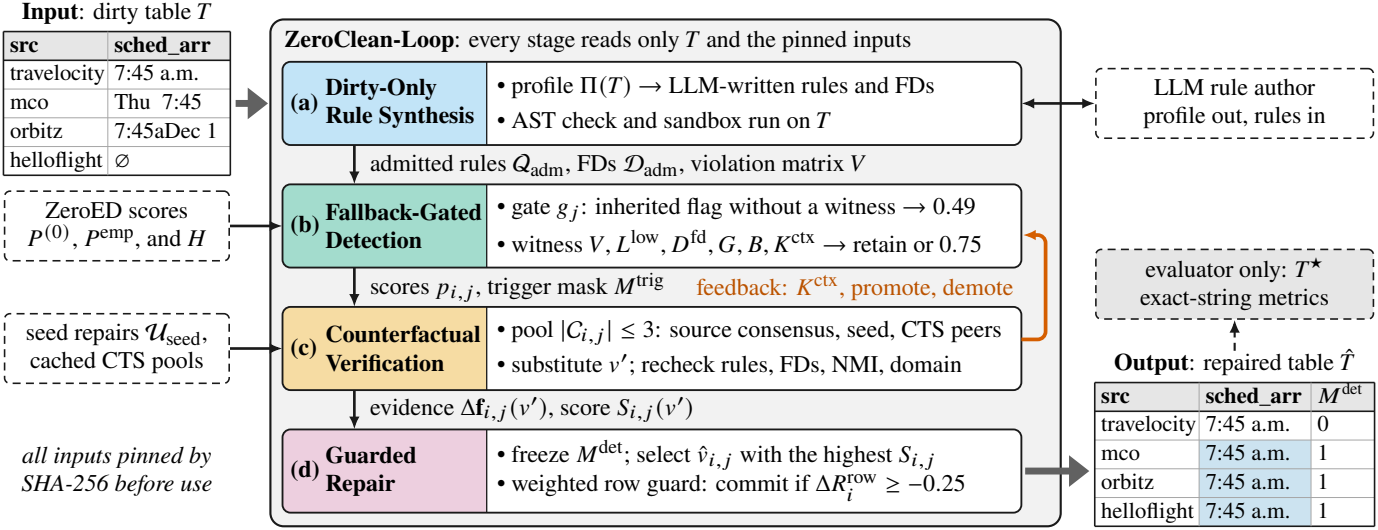
Input: dirty table T
src
sched_arr
travelocity
7:45 a.m.
mco
Thu 7:45
orbitz
7:45aDec 1
helloflight
∅
ZeroED scores $P^{(0)}$, P^{emp}, and H
seed repairs $\mathcal{U}_{\text{seed}}$, cached CTS pools
all inputs pinned by SHA-256 before use
ZeroClean-Loop: every stage reads only T and the pinned inputs
(a) Dirty-Only Rule Synthesis
• profile $\Pi(T)$ → LLM-written rules and FDs
• AST check and sandbox run on T
admitted rules Q_{adm}, FDs $\mathcal{D}_{\text{adm}}$, violation matrix V
(b) Fallback-Gated Detection
• gate g_j: inherited flag without a witness → 0.49
• witness $V, L^{\text{low}}, D^{\text{fd}}, G, B, K^{\text{ctx}}$ → retain or 0.75
scores $p_{i,j}$, trigger mask M^{trig}
feedback: K^{ctx}, promote, demote
(c) Counterfactual Verification
• pool $|C_{i,j}| \le 3$: source consensus, seed, CTS peers
• substitute v'; recheck rules, FDs, NMI, domain
evidence $\Delta\mathbf{f}_{i,j}(v')$, score $S_{i,j}(v')$
(d) Guarded Repair
• freeze M^{det}; select $\hat{v}_{i,j}$ with the highest $S_{i,j}$
• weighted row guard: commit if $\Delta R_i^{\text{row}} \ge -0.25$
LLM rule author
profile out, rules in
evaluator only: $T^\star$
exact-string metrics
Output: repaired table $\hat{T}$
src
sched_arr
M^{det}
travelocity
7:45 a.m.
0
mco
7:45 a.m.
1
orbitz
7:45 a.m.
1
helloflight
7:45 a.m.
1